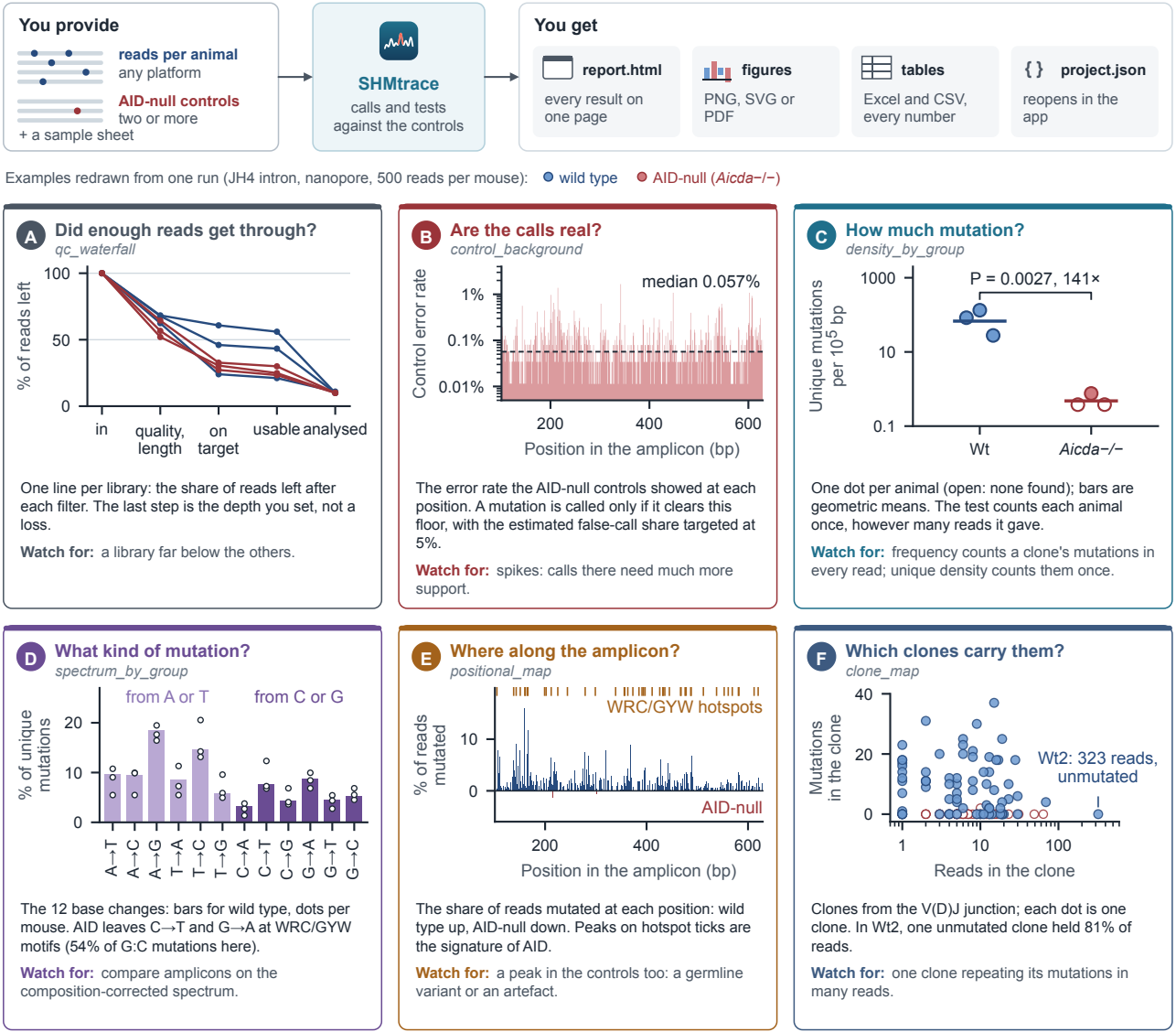

You provide
reads per animal
any platform
AID-null controls
two or more
+ a sample sheet
SHMtrace
calls and tests against the controls
You get
report.html
every result on one page
figures
PNG, SVG or PDF
tables
Excel and CSV, every number
{ } project.json
reopens in the app
Examples redrawn from one run (JH4 intron, nanopore, 500 reads per mouse): wild type AID-null (Aicda−/−)
A Did enough reads get through?
qc_waterfall
% of reads left
100
50
0
in
quality, length
on target
usable
analysed
One line per library: the share of reads left after each filter. The last step is the depth you set, not a loss.
Watch for: a library far below the others.
B Are the calls real?
control_background
Control error rate
1%
0.1%
0.01%
median 0.057%
200
400
600
Position in the amplicon (bp)
The error rate the AID-null controls showed at each position. A mutation is called only if it clears this floor, with the estimated false-call share targeted at 5%.
Watch for: spikes: calls there need much more support.
C How much mutation?
density_by_group
Unique mutations per 10^5 bp
1000
10
0.1
P = 0.0027, 141×
Wt
Aicda−/−
One dot per animal (open: none found); bars are geometric means. The test counts each animal once, however many reads it gave.
Watch for: frequency counts a clone's mutations in every read; unique density counts them once.
D What kind of mutation?
spectrum_by_group
% of unique mutations
from A or T
from C or G
20
10
0
A→T
A→C
A→G
T→A
T→C
T→G
C→A
C→T
C→G
G→A
G→T
G→C
The 12 base changes: bars for wild type, dots per mouse. AID leaves C→T and G→A at WRC/GYW motifs (54% of G:C mutations here).
Watch for: compare amplicons on the composition-corrected spectrum.
E Where along the amplicon?
positional_map
% of reads mutated
WRC/GYW hotspots
10
0
AID-null
200
400
600
Position in the amplicon (bp)
The share of reads mutated at each position: wild type up, AID-null down. Peaks on hotspot ticks are the signature of AID.
Watch for: a peak in the controls too: a germline variant or an artefact.
F Which clones carry them?
clone_map
Mutations in the clone
40
20
0
Wt2: 323 reads, unmutated
1
10
100
Reads in the clone
Clones from the V(D)J junction; each dot is one clone. In Wt2, one unmutated clone held 81% of reads.
Watch for: one clone repeating its mutations in many reads.